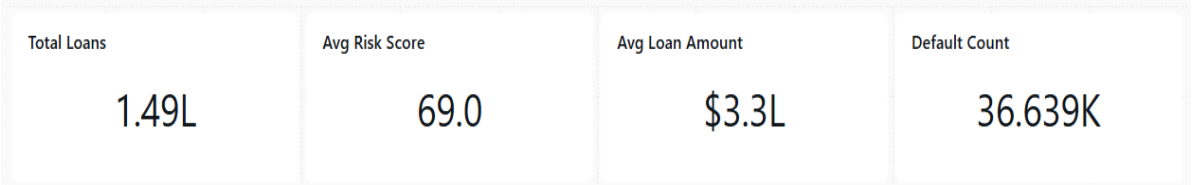
Credit Risk Assessment Pipeline Dashboards

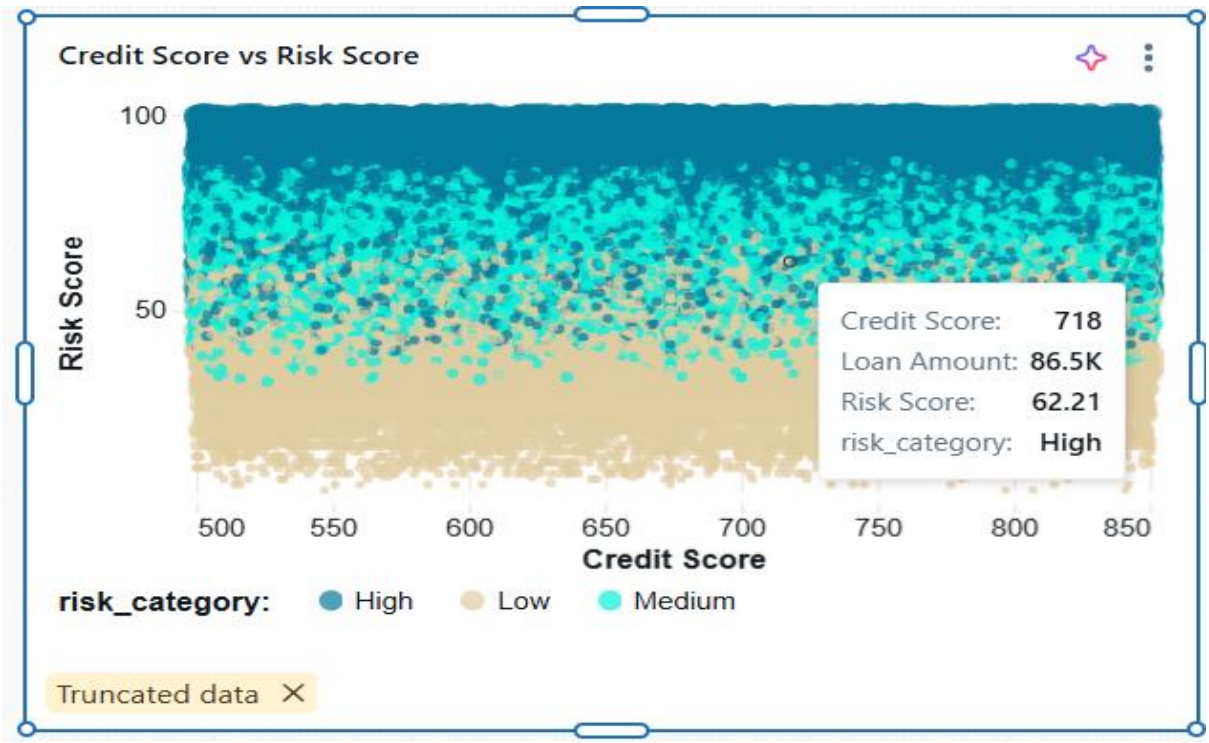
1. KPI Cards

KPI cards display key metrics such as total loans, average risk score, average loan amount, and total defaults. These metrics give a quick overview of the portfolio health. They help stakeholders understand overall trends without deep analysis. For example, a high average risk score may indicate increased exposure. These KPIs are essential for monitoring performance at a glance. They also support faster decision-making in dashboards.



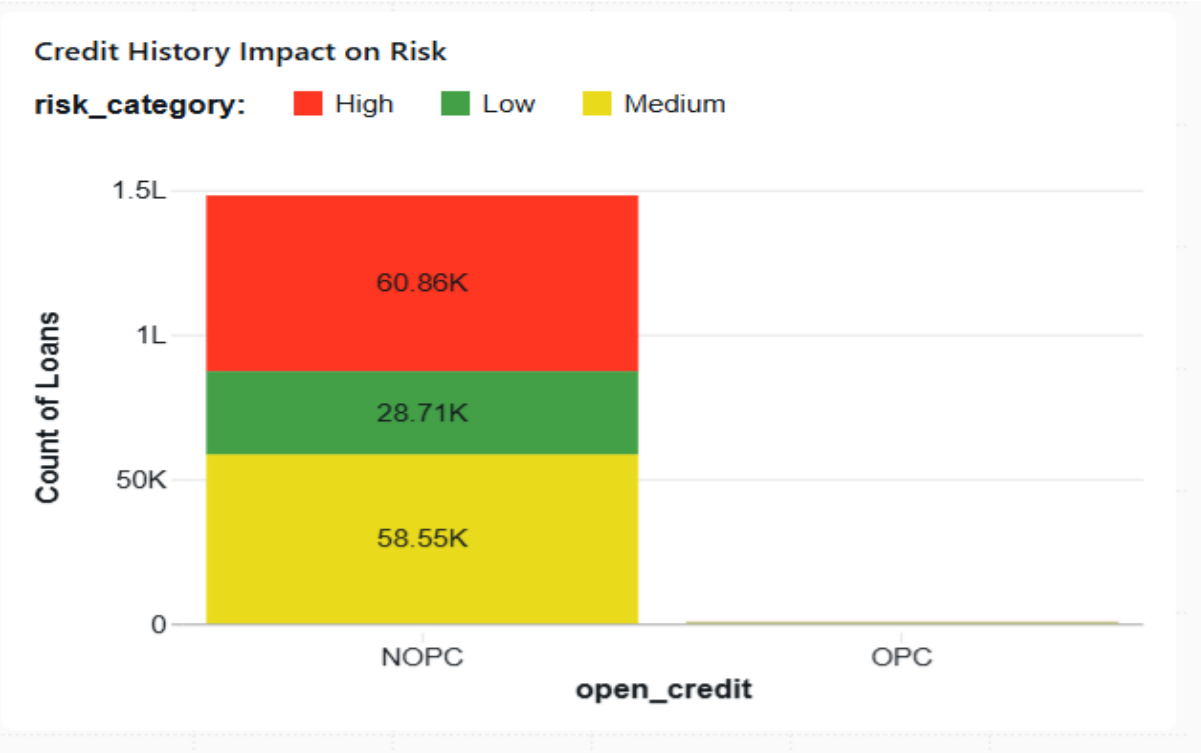
2. Loan Purpose Analysis

This chart analyzes loan distribution across different purposes along with associated risk categories. It helps determine which types of loans are more likely to be risky. For example, personal or emergency loans may show higher risk compared to home loans. Understanding this helps financial institutions tailor risk strategies for each loan purpose. It also assists in adjusting interest rates and approval criteria. This improves overall portfolio quality.



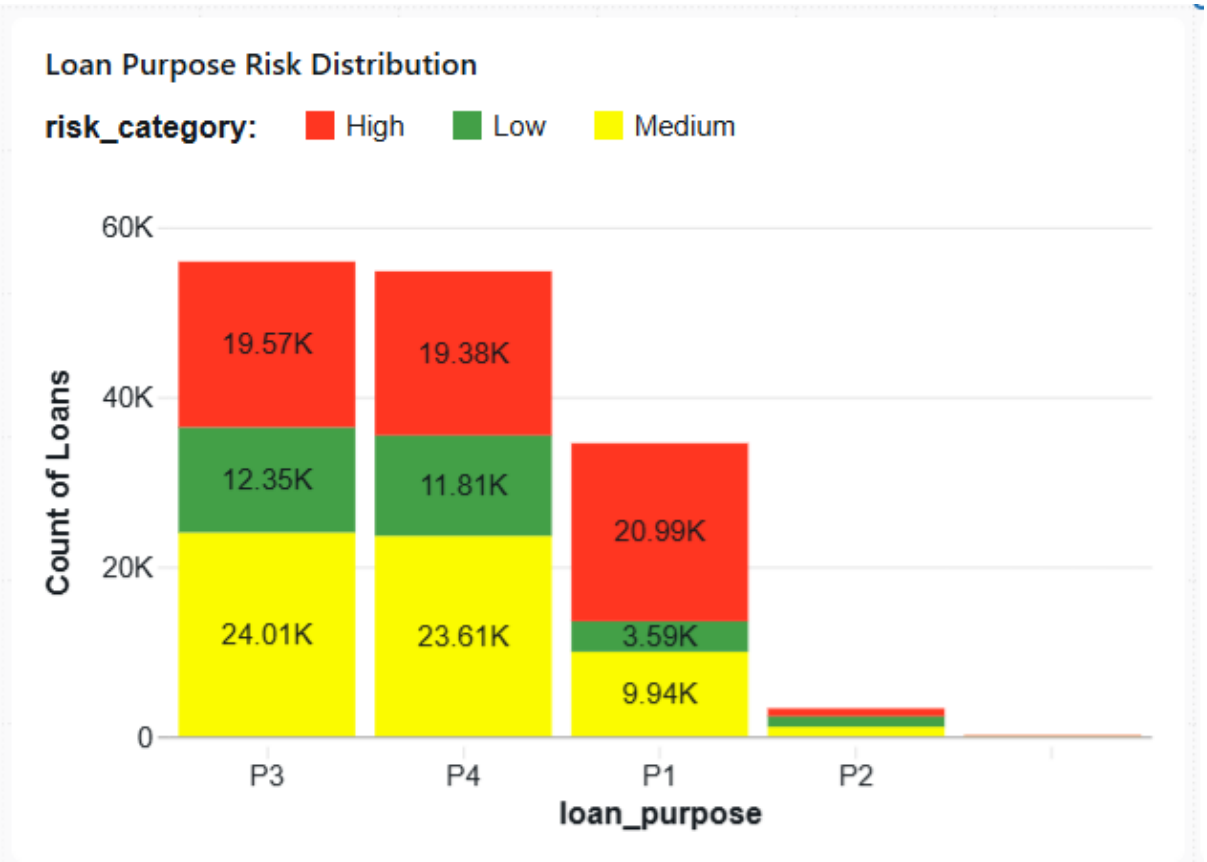
3. Credit History Impact

This visualization uses proxies like open_credit or debt-to-income ratio to analyze credit history impact. It helps understand how borrowing patterns relate to risk levels. A higher number of open credits or high debt ratios may indicate financial stress. By segmenting these factors with risk categories, lenders can detect risky credit behavior. This insight enhances creditworthiness evaluation. It also supports better loan approval decisions and risk control.



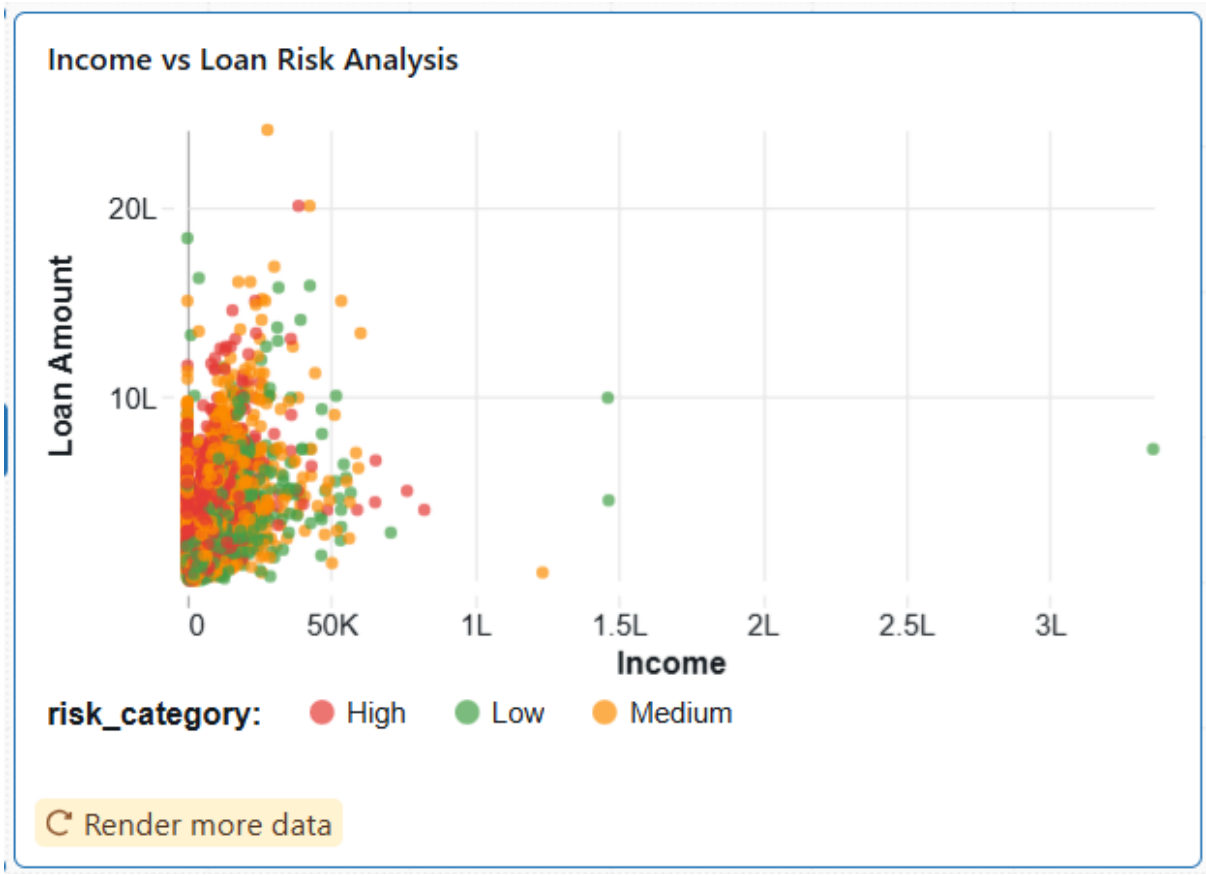
4. **Credit Score Impact**

This scatter plot visualizes how credit scores influence the calculated risk score of borrowers. Typically, lower credit scores are associated with higher risk scores. The plot helps identify patterns, trends, and outliers in borrower behavior. Adding loan amount as a size dimension provides additional context on exposure. This analysis is essential for validating the effectiveness of credit scoring models. It also helps lenders refine credit policies and scoring thresholds.



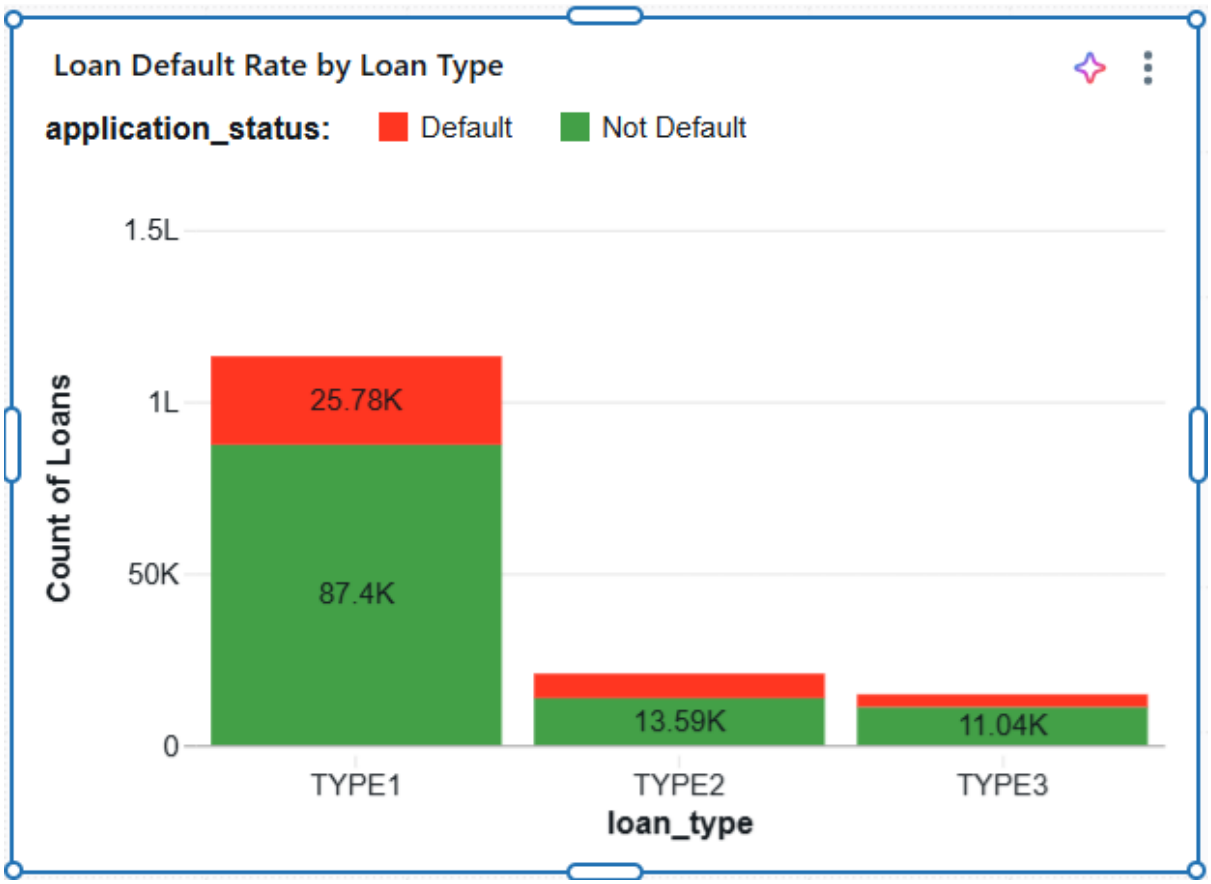
5. **Income vs Loan Risk**

This visualization compares borrower income with loan amount while highlighting risk categories. It helps determine whether higher-income individuals take safer or riskier loans. Patterns such as high loans with low income can indicate potential default risk. The color-coded risk categories make it easier to identify risky segments visually. This insight supports better financial profiling of borrowers. It also helps in setting income-based lending limits.



6. Loan Default Analysis

This chart compares loan defaults across various loan types using application status as a breakdown. It helps in identifying which loan categories are more prone to defaults. By analyzing this, lenders can understand risk patterns associated with different loan products. For example, unsecured loans may show higher default rates than secured ones. This insight is crucial for improving loan approval policies and risk assessment strategies. It also helps in optimizing product offerings.



7. **High-Risk Borrower Identification**

This visualization shows the distribution of borrowers across different risk categories such as High, Medium, and Low. It helps in quickly identifying the proportion of high-risk customers in the dataset. By applying filters like risk_score > 0.7, we can focus specifically on critical cases. This insight is useful for financial institutions to prioritize monitoring and mitigation strategies. A higher count in the high-risk category indicates potential default threats. It supports proactive decision-making in credit risk management.